

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 September 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	4 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

Sairunisha S M – Team Leader

Swetha R

Lokeshwari P

Yuvashri R

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	User registration through form; secure password hashing using bcrypt; user login using email and password; JWT token generation
FR-2	User Profile Management	View authenticated user profile; access profile using a valid JWT token; protect user-specific APIs
FR-3	Weather Information & AI Insights	Fetch real-time weather data; display temperature, humidity, wind and weather conditions; generate AI weather summary using Gemini; generate personalized recommendations for clothing, activities, travel and safety
FR-4	Favorite Location Management	Add/save favorite locations; view saved locations; update location details; delete saved locations

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The API should provide simple REST endpoints and clear JSON responses so that clients can easily request and understand weather information and AI recommendations.
NFR-2	Security	User passwords must be securely hashed using bcrypt, while JWT authentication must protect authorized endpoints and user-specific data.
NFR-3	Reliability	The system should handle invalid requests, authentication failures, unavailable locations, and external API errors gracefully and return appropriate responses.
NFR-4	Performance	API requests should be processed efficiently, with database operations and external weather/AI API calls performed only when required.
NFR-5	Availability	The backend should remain accessible whenever the Node.js server and required external services are available, with proper error handling for temporary service failures.